Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

Bab 1

PENDAHULUAN

The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

Bab 2

TINJAUAN PUSTAKA

Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Signed:
Date: 19 Pebruari 2025

Abstract

Abstract should be on one page. It should summarise the project objectives, the work carried out, methods used, the main research findings arising from the work and conclusions reached.

Acknowledgements

This thesis is the result of the authorâ€™s original research. However if any external support, help or contributions were received, it must be acknowledged here.

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[C]cSpeed of light in a vacuum, 299792458m/s [C]hPlank's constant, 6.62607004e-34$m^2.kg/s$[S]g$Gravitationalacceleratio$
kg/m^3

Bab 3

Introduction

There is no fixed page limit, but the thesis should reflect 14 weeks of full-time study. It should contain full details of the work completed. It should also, however, be concise and clear in its deliver. We would expect typical theses to have between 10,000 and 20,000 words. Please note, these are typical guidelines and penalty will not be applied for theses which are justifiably longer or shorter. Any additional material not suitable for the main text can be included in Appendices. Use Times New Roman, 12 point font, with 1.5 line spacing, fully justified.

The thesis structure and content should be discussed with the project supervisor, but would typically include an introduction, a section setting out objectives, scope and methodology, a critical literature review, research undertaken (experimental programme, data analysis and interpretation etc), finishing with conclusions and suggestions for further work. All sections must be numbered.

3.1 Format of Tables, Figures and Equations

Each item must have a numerical label and title (a caption). Figure captions must appear below the figure, see Fig. 3.1. Table captions must appear above the table, see Table 3.1. Equation captions appear to the right of the equation (or below if space does not permit), see Eq. eq:y.

A decaying harmonic is given as,

$$y(t) = ae^{-b\omega t}\sin(\omega t + \phi) \tag{3.1}$$

where a is the sine wave amplitude, frequency ω, phase offset ϕ, and the exponential coefficient $b > 0$. An example is shown in Figure 3.1.

This is generally an infinite dimensional optimisation problem of the form

$$\min_{u,x} \ \phi\left(x(t_f)\right) \ + \int_{t_0}^{t_f} f_0\left(x(t), u(t)\right)\, dt \tag{3.2}$$

subject to $\dot{x}(t) = f\left(x(t), u(t)\right)$
$c\left(x(t), u(t)\right) \leq 0, \ \ t \in [t_0, t_f]$
$\omega\left(x(t_0), x(t_f)\right) = 0$ where $f(x(t), u(t))$ describes the dynamic of the system, the function $\omega(x(t_0), x(t_f))$ the equality constraints on initial and final states and $c(x(t), u(t))$ the path constraints. The solution of an optimal control problem is the pair $[x^*(t), u^*(t)]$ which minimise the objective function defined by $\phi(x(t_f))$ and $f_0(x(t), u(t))$.

Tables, Figures and Equations must have separate numbering. Each item should appear in sequential order and be mentioned explicitly in the body of text, and be mentioned prior to appearing in the report. If you are using a figure or table from a third party (book, paper, website etc.) then it must be referenced in the caption, as in Tables 3.1–3.2. If data has been extracted from several sources and compiled into a table, this can be referenced in the table itself.

Each item should be centralised in the body of text. No text wrapping should be employed around figures, tables, or equations. All text in the figure should be legible. There is no limitation of figure size, it is better to show a figure larger and legible, than smaller and indecipherable.

Day	Min Temp	Max Temp	Summary
Monday	11	22	A clear day with lots of sunshine. However, the strong breeze will bring down the temperatures.
Tuesday	9	19	Cloudy with rain, across many northern regions. Clear spells across most of Scotland and Northern Ireland, but rain reaching the far northwest.
Wednesday	10	21	Rain will still linger for the morning. Conditions will improve by early afternoon and continue throughout the evening.

Tabel 3.1: Table caption website:table

	Item	
(r)1-2 Animal	Description	Price ($)
Gnat	per gram	13.65
	each	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Tabel 3.2: Caption for table taken from [**?**]

3.2 Nomenclature

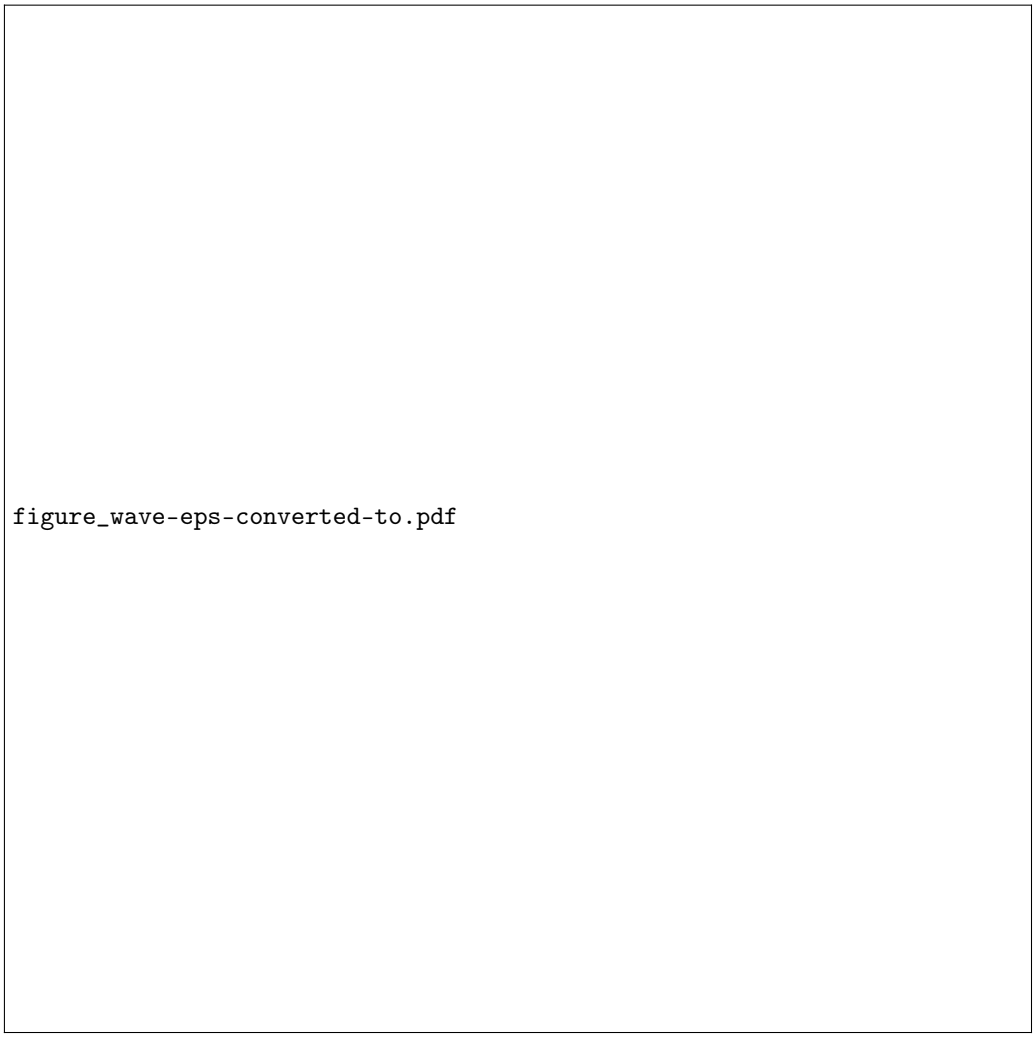
All variables and symbols used should be given in the text, and summarised in List of Symbols, or Nomenclature at the start. The nomenclature should include all the variables, their descriptions and if applicable, the values (in the case of constants) and units. It does not usually include acronyms.

The symbols can be grouped, or not, to help with clarity depending on the number of variables.

3.3 Referencing

Referencing should preferably follow the Harvard system, although a numeric style such as Vancouver system is also acceptable. It is important that attention is paid to proper referencing of others' work (including figures), both to avoid plagiarism and to allow others to find the source document blackholes, e.g., web links should contain the URL and the date that the web link was sourced. If necessary, Turnitin will be used to check for plagiarism.

A bibliography can be used to reference different reference sources including books book:goossens93, and articles art:AbedonHymanThomas2003,art:greenwade93. proc:Abedon1994 wrote an interesting paper in the field of molecular biology, which has a slightly different citation format.



Gambar 3.1: Exponentially decaying sine wave from Eq. eq:y

Bab 4

Research and Results

4.1 Section 1

4.2 Section 2

4.2.1 Sub-section

4.3 Section 3

Bab 5

Conclusion

This section should summarise the information presented in the report. No new information regarding the work should be included here. It will often include a further or future work section, discussing what would be useful going forward.

Lampiran A

Additional material

Appendices give the chance to include supplementary information which is useful but not pertinent to the understanding of the report. Appendices should be referred to explicitly in the main text, or should not be included. The appendices shall not be used as a means to include additional word count.